

malicious files. Web robots make use of instant messaging, Internet Relay Chat or other Web interfaces to communicate with other users of Internet-based services. Internet Relay Chats also provide help services for any new user by lurking in the background of any conversation channel and commenting on some of the phrases uttered by users (based on pattern matching).

Different types of Web bots and how they are used

Web bots can be used to perform both legitimate and malicious tasks. Depending on the application for which a Web bot is used, it is classified into the following categories.

Legitimate Web robots: These perform legitimate actions on a Web application. They save a lot of time and effort for users. If the same task was performed manually, then it would have taken much longer time. These types of Web bots actually serve a purpose and act as time savers for different applications. They are broadly categorised as follows:

1. **Spider Web robots:** These are used by different search engines to explore different Web pages for organisation, content and linking. They have got certain criteria for indexing, based on which they determine the ranking of different Web pages within the search results. Such types of Web robots are used by Google and are called Googlebots.
2. **Trading Web robots:** These traverse or 'spider' different online auction sites in order to locate the best deals on any specific product or service. In such scenarios, a trading Web robot is used for commercial gain. These robots are also helpful in doing a complete analysis of different online commercial applications and provide users the best deals. Trivago, an online hotel booking Web application, uses such bots to display the best hotel deals.
3. **Media Web robots:** These are used to provide the latest news updates, weather conditions, sports news, currency exchange rates, etc. They are also used as censors in different applications that run instant messenger programs and chat rooms. Such Web bots are widely used by different online messenger applications like eBuddy, IMO, etc.

Malicious Web robots: Malicious Web robots are a collection of such bots, used in different ways to perform some malicious act or network security breaches. Such bots are widely used by hackers all across the globe.

1. **Spam Web robots:** These collect different data from various forms that are filled online. They are also used for spreading advertisements with the help of pop-ups. Different advertisement firms also use them for collecting email addresses of people to spam them with advertisements.
2. **Hacker Web robots:** Hackers use these to crawl around the Internet and thus find the vulnerabilities in different websites and online applications so that they can be exploited for malicious purposes.
3. **Botnets:** These are actually the networks that different

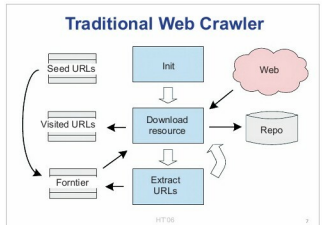


Figure 1: Flow diagram of a traditional Web crawler (Source: googleimages.com)

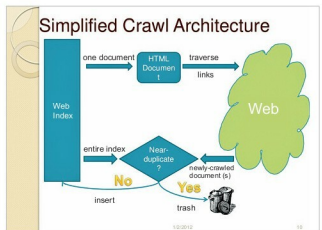


Figure 2: Architecture of a Web crawler (Source: googleimages.com)

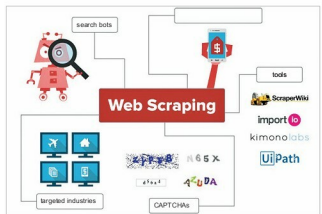


Figure 3: Web scrapping done by malicious Web robots (Source: googleimages.com)

hackers set up online by using zombie computers in order to perform various malicious acts such as the denial of service attacks. Zombie computers are those that have been taken over by a hacker without the owner's knowledge.

4. **Download Web robots:** These are used to forcibly download any specific Web page that the hacker wants the surfer to see instead of the one that the surfer requested.